

Silverline House — 25-Storey Residential HRB

MEP Sprinkler Design Specification

Part B — Fire Safety | BS EN 12845 / LPCB

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1 Scope and Design Standards

This specification covers the design, installation, and commissioning of the automatic wet pipe sprinkler system for Silverline House. The system has been designed in accordance with BS EN 12845:2015 (Fixed firefighting systems — Automatic sprinkler systems) and the Loss Prevention Certification Board (LPCB) technical requirements. The designer is LPCB-certificated to BS EN ISO 9001:2015 for the design of automatic sprinkler systems. The design has been developed in coordination with the Fire Strategy Report (document ref 2.2.1) and the Mechanical Services Coordination Drawings.

The system classification adopted is Ordinary Hazard Group 2 (OH2) for all residential floors and the retail unit, and Extra Hazard Group 1 (EH1) for the basement car parking levels and plant rooms, in accordance with BS EN 12845 Table 2. Design parameters: density 5 mm/min over a design area of 144 m² for OH2; density 10 mm/min over 260 m² for EH1.

2.1 Head Specification

High response sprinkler heads are specified throughout with a rated temperature of 93°C in accordance with BS EN 12259-1. The heads are selected from an LPCB-certificated range and comply with the listings requirements of FM Global Property Loss Prevention Data Sheet 2-0. The response time index (RTI) of the specified heads is $\leq 50 \text{ (m}\cdot\text{s)}^{1/2}$, classifying them as standard response in accordance with BS EN 12845 Table 4.

Head positions are coordinated with the architectural reflected ceiling plans (ref SLH-ARC-RC-001 to 048) to achieve the maximum spacing and area of coverage specified in BS EN 12845 Table 7. Heads in the residential areas are concealed pattern with a chrome finish to suit the interior design. Heads in plant rooms and the car park are upright pattern with unfinished brass bodies.

The water demand for the sprinkler system is 1,800 l/min for the most hydraulically onerous design area, sustained for a minimum period of 60 minutes in accordance with BS EN 12845. The calculation assumes a minimum inlet pressure of 0.3 bar at the base of the system riser.

2.2 Pipe Sizing and Layout

Distribution pipework is in Schedule 10 black steel to BS EN 10255 with grooved mechanical couplings. Branch pipework is in CPVC (chlorinated polyvinyl chloride) to FM Approval Standard 1635 where concealed in ceiling voids within the residential areas. The system is arranged in a tree-and-branch configuration with a main distribution ring main at each plant room level feeding risers to each zone. Hydraulic calculations have been carried out in accordance with BS EN 12845 Annex A using specialist hydraulic calculation software. Minimum pipe diameter is 25 mm (NB) for branch pipes and 100 mm for main risers.

Active fire suppression systems installed in the building comprise a wet pipe sprinkler system covering all residential floors, plant rooms, and car parking basement levels; dry riser inlets at the external face of the building for fire service use; and a wet riser system with outlets on each residential floor. All suppression systems have been designed by a specialist MEP engineer with LPCB certification and comply with the relevant BS EN and Loss Prevention Standards. Design calculations and system schematics are contained in the MEP Sprinkler Design Specification document.